

The image displays a series of Feynman diagrams representing a scattering process, likely in a gauge theory. The diagrams are organized into four terms, separated by plus signs, and followed by an integral expression.

Diagram 1 (Leftmost): A box diagram with two vertices. The left side has two external lines with momenta p_1 and p_2 . The right side has two external lines with momenta p_1 and p_2 . A wavy line connects the two vertices, with momentum k flowing from left to right. A dashed red line is vertical, and a dashed blue line is diagonal. The diagram is labeled with $p_{12} - k$ on the right side.

Diagram 2: A box diagram with two vertices. The left side has two external lines with momenta k and $k - p_1$. The right side has two external lines with momenta $k - p_1$ and $k - p_2$. A wavy line connects the two vertices, with momentum k flowing from left to right. A dashed red line is vertical, and a dashed blue line is diagonal. The diagram is labeled with $p_{12} - k$ on the right side.

Diagram 3: A box diagram with two vertices. The left side has two external lines with momenta p_1 and $k - p_2$. The right side has two external lines with momenta $k - p_2$ and $k - p_1$. A wavy line connects the two vertices, with momentum k flowing from left to right. A dashed red line is vertical, and a dashed blue line is diagonal. The diagram is labeled with $p_{12} - k$ on the right side.

Diagram 4 (Rightmost): A box diagram with two vertices. The left side has two external lines with momenta k and $k - p_1$. The right side has two external lines with momenta $k - p_1$ and $k - p_2$. A wavy line connects the two vertices, with momentum k flowing from left to right. A dashed red line is vertical, and a dashed blue line is diagonal. The diagram is labeled with $p_{12} - k$ on the right side.

The diagrams are summed and followed by the integral expression:

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\boxtimes}^{qq'} \delta_{q^2}^+(p_{12} - k) \delta_0^+(k)}{p_1^2 p_2^2 (k - p_2)^2 (k - p_1)^2}$$